Roman Šafranko

Junior Software Engineer

roman.safranko@outlook.com • +421 949 668 401 • Prešov, Slovakia • romansafranko.com • github.com/Mekr0t

PROFILE

Computer science graduate focused on full-stack web development and applied machine learning. Experienced with React and Next.js, Python, PostgreSQL and Docker through self-directed projects taken from concept to deployment.

PROJECTS

Prismatic Arena 2026

- Teamfight Tactics statistics platform on Next.js and the Riot Games API, backed by PostgreSQL and Redis in Docker.
- Designed an automated labelling pipeline clustering raw board states into named compositions, replacing hand-maintained tier lists.

VB Visualizer 2025 – 2026

- Browser music visualiser producing 9:16 video for short-form platforms, vanilla ES modules, Web Audio API, Canvas. No build step, no backend.
- Implemented spectral-flux onset detection for beat-synced animation and LRCLIB integration for automatic lyric timing.

Voice-controlled robotic arm 2026

- Speech to local LLM to motion planning on a Dobot Magician, with OpenCV closing the loop on object localisation.
- Ran the full pipeline locally, speech recognition, inference and vision, with no cloud API.

nullres 2026

Purged walk-forward backtester

- Walk-forward backtesting framework in Python/pandas for evaluating trading signals under strict leakage control.
- Implemented purged walk-forward cross-validation, running each strategy with and without leakage to quantify lookahead-bias inflation.
- Documented every disproved hypothesis with its metrics rather than only the surviving ones.

CV Studio 2026

CV builder with ATS mode

- React CV builder with ATS-safe templates, content linting, job-description keyword matching and multi-format export.
- Runs entirely in the browser; this CV was generated with it.

Windows provisioning tool 2026

- PowerShell toolkit that provisions a fresh Windows install in one pass: 35 apps, 16 tweaks, 7 scripts, 4 presets.

EDUCATION

Ing. Artificial Intelligence 2026 – 2028 (expected)

Technical University of Košice · Košice

B.Sc. Computer Science 2023 – 2026

Technical University of Košice · Košice

Thesis: Controlling the Dobot Magician robotic arm using multimodal language models.

SKILLS

JavaScript / TypeScript · React & Next.js · Node.js · Python · SQL & PostgreSQL · Redis · Linux · Computer vision · LLM integration

LANGUAGES

Slovak — Native **English** — C1